

1. A 57-year-old woman comes to the physician because of a 3-day history of fever, rash on her chest and back, and diffuse body aches. She has a seizure disorder and has been taking phenytoin for 1 month. She also has hypertension treated with hydrochlorothiazide. She has no known allergies. She has smoked one-half pack of cigarettes daily for 40 years and drinks one glass of wine daily. Her temperature is 38.4°C (101.1°F), pulse is 70/min, and blood pressure is 130/70 mm Hg. Pulse oximetry on room air shows an oxygen saturation of 98%. Soft mobile lymph nodes are palpated in the cervical, axillary, and inguinal chains; they measure less than 1 cm each. There is an erythematous maculopapular rash over the trunk. The abdomen is tender in the right upper quadrant, and the liver edge is palpated 3 cm below the right costal margin. Laboratory studies show:

Hematocrit	40%
Leukocyte count	10,000/mm ³
Segmented neutrophils	55%
Eosinophils	10%
Lymphocytes	30%
Monocytes	5%
Platelet count	250,000/mm ³
Serum	
Creatinine	1 mg/dL
AST	150 U/L
ALT	170 U/L
Hepatitis C antibody	negative
HBsAg	negative
Anti-HBc	negative
Anti-HBsAg	negative

Blood cultures are negative. Which of the following is the most appropriate next step in management?

- ☐ A) Measurement of serum α -fetoprotein concentration
- ☐ B) Serum parvovirus antibody assay

Blood cultures are negative. Which of the following is the most appropriate next step in management?

- A) Measurement of serum α -fetoprotein concentration
- B) Serum parvovirus antibody assay
- C) Discontinuation of phenytoin therapy
- D) Prednisone therapy
- E) Switch from hydrochlorothiazide to lisinopril
- F) Bone marrow biopsy
- G) Lymph node biopsy

2. A 20-year-old man is brought to the emergency department 30 minutes after the sudden onset of severe headache followed by loss of consciousness for 10 minutes. Yesterday, he tripped and fell down one flight of stairs. He was dazed for approximately 5 minutes after the fall but was otherwise uninjured. He has had fever, nasal congestion, and cough for 4 days treated with aspirin and pseudoephedrine. One week ago, he had a severe headache that persisted for 1 hour and was relieved with aspirin. On arrival, he is stuporous. His temperature is 37.8°C (100°F), pulse is 100/min, respirations are 14/min, and blood pressure is 180/90 mm Hg. The pupils are round and reactive to light. Ocular movements are full. Funduscopic examination shows no abnormalities. There is moderate nuchal rigidity with forward flexion. Muscle tone is normal. Deep tendon reflexes are increased but symmetrical. Babinski sign is present bilaterally. The patient moves his extremities symmetrically in response to painful stimuli. A CT scan of the head without contrast is shown. Which of the following is the most likely cause of this patient's loss of consciousness?

- A) Bacterial meningitis
- B) Head trauma
- C) Herpes simplex encephalitis
- D) Ruptured berry aneurysm
- E) Tuberculous meningitis



The response options for the next 2 items are the same. Select one answer for each item in the set.

For each patient with vertigo, select the most likely diagnosis.

- A) Acoustic schwannoma
- B) Benign positional vertigo
- C) Cerebellar hemorrhage
- D) Complex partial seizure
- E) Endolymphatic fistula
- F) Hyperventilation syndrome
- G) Meniere disease
- H) Postural hypotension
- I) Vertebrobasilar transient ischemic attack

3. A 50-year-old man has had episodes of sudden severe dizziness over the past 3 years. He describes these episodes as a sensation of spinning. They are associated with sweating, pallor, nausea, and occasional vomiting. He has not noted an obvious precipitant. The episodes occur as often as weekly and as infrequently as every 3 to 4 months. They usually last 2 to 3 hours. He has noticed a decrease in hearing and a constant hissing noise in the right ear. Examination shows hearing loss in the right ear.

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4. Over the past 2 years, a 35-year-old woman has had episodes of a spinning sensation. The episodes occur abruptly, are associated with nausea and vomiting, and last about 1 hour. The first episode occurred when the airplane in which she was traveling ascended to cruising altitude. Subsequent episodes have occurred on elevators, and she has had mild episodes when she sneezes. Two years ago, she sustained a concussion after a fall. Examination shows decreased hearing in the right ear.

5. A 5-year-old boy with cerebral palsy is brought to the physician for a school physical examination. He did not walk until the age of 3 years and is just learning to use utensils to feed himself. He wears hearing aids because of bilateral sensorineural hearing loss. He receives physical, occupational, and speech therapy. He is at the 5th percentile for height and weight; he has remained below the 5th percentile for head circumference since birth. Examination shows right esotropia. There is increased muscle tone in all extremities, bilateral ankle spasticity, and diffuse hyperreflexia. He has an ataxic gait. Psychometric testing shows an IQ of 65. A CT scan of the head shows punctate hyperintensities along the ventricular margins. Which of the following is the most likely underlying cause of this child's abnormalities?
- A) Chronic lead exposure
 - B) Congenital cytomegalovirus infection
 - C) Fragile X syndrome
 - D) Phenylketonuria
 - E) Tuberous sclerosis

6. Three days after admission to the hospital because of an 8-hour history of weakness and numbness of the right side of the face and right upper extremities, a 32-year-old woman becomes comatose. On admission, she was treated with aspirin. She has fibromuscular dysplasia. Her pulse is 40/min, and blood pressure is 160/100 mm Hg. She is not responsive to verbal or tactile stimuli. Examination shows an 8-mm left pupil that is nonreactive and a 3-mm right pupil that is reactive. Her left eye is deviated outward. There is occasional movement of the left hand and foot and no movement of the right side of the face or right extremities. Deep tendon reflexes are 4+ in the right upper and lower extremities. Babinski sign is present on the right. Her serum sodium concentration is 131 mEq/L. Which of the following is the most likely diagnosis?

- ☐ A) Cerebellar hematoma
- ☐ B) Metabolic encephalopathy
- ☐ C) Midbrain hemorrhage
- ☐ D) Pontine infarction
- ☐ E) Uncal herniation

7. A 47-year-old woman comes to the physician because of a 4-month history of difficulty walking, constant headache, and urinary incontinence. Her symptoms have worsened over the past month. Treatment with acetaminophen and ibuprofen has not relieved her headache. She has had mild memory impairment since a motor vehicle collision 5 years ago in which she lost consciousness. She has had two generalized tonic-clonic seizures over the past year. Her sister has multiple sclerosis. The patient's vital signs are within normal limits. Examination shows normal strength in the upper and lower extremities and increased muscle tone in the lower extremities. Deep tendon reflexes are normal in the upper extremities and brisk in the lower extremities. Babinski sign is present bilaterally. Her gait is broad-based and spastic. Sensation is intact. Which of the following is the most likely location of this patient's lesion?

- A) Cauda equina
- B) Cervical spinal cord
- C) Parasagittal cerebral cortex
- D) Pons
- E) Thoracic spinal cord



8. A 47-year-old man comes to the physician because of a 3-month history of intermittent pain of the right side of his face. The pain begins in the region of the ear, radiates to the chin, and lasts for 1 to 2 seconds. It is precipitated by exposure to cold weather, drinking a cold liquid, eating, and brushing his teeth. He has hypertension and gastroesophageal reflux disease. Medications include hydrochlorothiazide and antacids. Vital signs are within normal limits. Examination shows no abnormalities. Which of the following is the most appropriate pharmacotherapy for pain in this patient?

- A) Capsaicin
- B) Carbamazepine
- C) Diclofenac
- D) Fluoxetine
- E) Oxycodone

9. A 67-year-old man is brought to the physician by his wife because of a 6-month history of difficulty remembering both recent and remote events. He also has had jerky movements of his arms and legs for 4 weeks. He has had loss of balance and has fallen several times while walking. He is inattentive. Examination shows frequent myoclonic jerking of all extremities, increased muscle tone, and hyperactive reflexes. Babinski sign is present bilaterally. A CT scan of the head shows mild atrophy. Cerebrospinal fluid analysis shows no abnormalities except for a protein concentration of 80 mg/dL. EEG shows a generalized slow background interrupted by synchronous sharp wave complexes. Which of the following is the most likely diagnosis?

- A) Creutzfeldt-Jakob disease
- B) Dementia, Alzheimer type
- C) Dementia with Lewy bodies
- D) Hepatolenticular degeneration (Wilson disease)
- E) Multi-infarct (vascular) dementia



10. A 63-year-old man is admitted to the hospital for a lumbar laminectomy. One day postoperatively, he has a low-grade fever secondary to a urinary tract infection. He also has difficulty sleeping at night, and treatment with lorazepam is started. Two days postoperatively, he becomes drowsy and confused. On examination, he is difficult to arouse, and his speech is slurred. He is able to answer a few simple questions, but he thinks that he is at home and that the year is 1956. He cooperates poorly with formal strength testing. The remainder of the examination shows no abnormalities. Which of the following is the most likely diagnosis?

- A) Cerebral infarction
- B) Dementia, Alzheimer type
- C) Metabolic encephalopathy
- D) Nonconvulsive status epilepticus
- E) Subdural hematoma

11. A 12-year-old girl is brought to the physician because of frequent headaches over the past 2 months and vomiting and nausea for 2 days. Her chiropractic physician advised her parents to seek additional evaluation because she has not responded to manipulation and massage therapy. The patient takes no prescription medications, but the entire family takes multiple supplements including megadoses of multivitamins. Her mother had a stomach virus 1 week ago. The patient's blood pressure is 130/80 mm Hg. Neurologic examination shows no abnormalities. A photograph of the optic fundus is shown. Laboratory studies show:

Serum	
Na ⁺	140 mEq/L
K ⁺	4.2 mEq/L
Prolactin	20 ng/mL
Urine specific gravity	1.030



Which of the following is the most likely cause of this patient's optic findings?

- A) Diabetes insipidus
- B) Frontal lobe tumor
- C) Pituitary microadenoma
- D) Systemic lupus erythematosus
- E) Vitamin A toxicity

12. A 77-year-old man is brought to the emergency department 3 hours after the acute onset of **dizziness and unsteadiness** on awakening this morning. He has also noticed difficulty swallowing water. He has hypertension treated with atenolol. He has vomited once in the emergency department. His temperature is 37°C (98.6°F), pulse is 65/min, respirations are 20/min, and blood pressure is 170/100 mm Hg. Examination shows left ptosis. The left pupil is 2 mm, and the right pupil is 5 mm. Both pupils are reactive to light. There is coarse nystagmus on left lateral gaze. Muscle strength is normal. Sensation to temperature is decreased over the left side of the face and the right side of the body. The most likely cause of these findings is occlusion of which of the following arteries?

- A) Left middle cerebral
- B) Left posterior inferior cerebellar
- C) Left vertebral
- D) Right middle cerebral
- E) Right posterior inferior cerebellar
- F) Right vertebral

13. A 32-year-old man is brought to the emergency department by a friend because of a 1-week history of increasingly bizarre behavior. Earlier this evening, the patient started to drive at 100 miles per hour, weaving in and out of traffic on the interstate highway, until one of his passengers convinced him to drop her off at her apartment. His friend then insisted on bringing him to the hospital for evaluation. The patient works at a brokerage firm and has been at the office all night on several occasions this past week. Several of his customers have started to complain that he calls late at night and talks about plans for their money that do not make sense. His friend describes this behavior as completely uncharacteristic because he was previously a successful money manager. The patient typically drinks four beers on weeknights and up to a dozen on weekends. His temperature is 37°C (98.6°F), pulse is 95/min, respirations are 10/min, and blood pressure is 140/90 mm Hg. Physical examination shows no other abnormalities. On mental status examination, his affect is inappropriately bright. He speaks quickly and is difficult to interrupt. He says that he is upset about being brought to the emergency department; he then laughs and states that he can clear everything up. He says that he has received a secret message regarding a new economic development that will enable all of his customers and him to become wealthy. He has been staying up nights charting it all out and now knows exactly what needs to be done. Urine toxicology screening is negative. Which of the following is the most likely diagnosis?

- A) Attention-deficit/hyperactivity disorder
- B) Bipolar disorder
- C) Cyclothymic disorder
- D) Delusional disorder
- E) Major depressive disorder with psychotic features
- F) Schizoaffective disorder
- G) Schizophrenia
- ☒ H) Substance-induced mood disorder



14. An 82-year-old woman comes to the physician because of a 1-year history of neck stiffness, pain radiating from her neck to her left hand, urinary incontinence, and unsteadiness when walking. Medical history includes osteoarthritis, peptic ulcer disease, and hyperlipidemia. Current medications are naproxen, ranitidine, and lovastatin. She has a history of alcohol abuse but has abstained from drinking for 5 years. Neurologic examination shows atrophy of the left upper extremity and increased tone in both lower extremities. Muscle strength in the left upper extremity is 3/5. Deep tendon reflexes are 2+ at the biceps and brachioradialis bilaterally, absent at the left triceps, and 3+ at the knees and ankles. Babinski sign is present bilaterally. Sensation to pinprick and temperature is decreased over the left dorsal forearm and hand. Proprioception and sensation to vibration are markedly decreased at the ankles and toes. She has a stiff, wide-based gait. Which of the following is the most likely diagnosis?

- A) Bilateral brachial plexopathy
- B) Epidural cord compression
- C) Parasagittal meningioma
- D) Peripheral neuropathy
- E) Spondylotic cervical myelopathy

15. A 37-year-old man with a 10-year history of AIDS comes to the emergency department because of a 3-day history of diffuse headache and mild weakness of his left hand. He has not had fever or chills. He has been unable to tolerate highly active antiretroviral therapy due to severe nausea. His only medication is trimethoprim-sulfamethoxazole. His most recent CD4+ T-lymphocyte count 2 weeks ago was 50/mm³ (Normal ≥500), and plasma HIV viral load was 100,000 copies/mL. He is alert and oriented to person, place, and time. His pulse is 100/min, and blood pressure is 90/60 mm Hg. Muscle strength is 4/5 in the left hand. Neurologic examination shows no other focal findings. Laboratory studies show:

Serum	
Na ⁺	128 mEq/L
K ⁺	3.8 mEq/L
Cl ⁻	93 mEq/L
HCO ₃ ⁻	25 mEq/L
Creatinine	1 mg/dL
Urine	
Sodium	80 mEq/L
Osmolality	500 mOsmol/kg H ₂ O

A CT scan of the head shows a ring-enhancing lesion in the right parietal cortex. Which of the following is the most likely explanation for these laboratory findings?

- A) Acute tubular necrosis
- B) Addison disease
- C) Hypovolemia
- D) Salt-wasting nephropathy
- E) Syndrome of inappropriate secretion of ADH (vasopressin)

16. A previously healthy 21-year-old man is brought to the physician 1 hour after sustaining an injury to his right shoulder during a college football game. Examination shows a bulge anterior to the shoulder. An x-ray of the right shoulder shows an anterior dislocation. After closed reduction, there is numbness over the proximal lateral arm and marked weakness of abduction. Which of the following is the most likely site of nerve injury?

- ☐ A) Axillary nerve
- ☐ B) Cervical nerve root at the cervical foramen
- ☐ C) Long thoracic nerve
- ☐ D) Median nerve above the elbow
- ☐ E) Median nerve at the elbow
- ☐ F) Median nerve at the wrist
- ☐ G) Musculocutaneous nerve above the elbow
- ☐ H) Musculocutaneous nerve at the elbow
- ☐ I) Musculocutaneous nerve at the wrist
- ☐ J) Radial nerve above the elbow
- ☐ K) Radial nerve at the elbow
- ☐ L) Radial nerve at the wrist
- ☐ M) Suprascapular nerve
- ☐ N) Thoracodorsal nerve
- ☐ O) Ulnar nerve at the elbow



17. A 26-month-old boy is brought to the physician for a follow-up examination 2 weeks after completing a 10-day course of amoxicillin for an ear infection. His father is concerned because his son has had many ear infections since birth, and his speech is difficult to understand. The child walks and runs with only occasional falls and is able to climb on furniture. He can stack seven blocks without difficulty. He is at the 10th percentile for height and weight. During the examination, he does not say any intelligible words, and he intermittently babbles or screams. Examination shows retracted tympanic membranes with poor movement with pneumatic otoscopy. Which of the following is the most appropriate next step in management?

- A) Reassurance
- B) Measurement of blood lead concentration
- C) Audiometry
- D) CT scan of the head
- E) 10-Day course of high-dose amoxicillin-clavulanate
- F) Insertion of tympanostomy tubes

18. A 42-year-old man comes to the physician because of a 6-month history of urinary urgency and incontinence. He says that he feels the urge to void small volumes of urine every 2 hours. Occasionally, he has been unable to reach the bathroom in time. Six months ago, he sustained a fracture of C6 when he dove into a shallow pool. He had quadriparesis with loss of sensation to pinprick below C6. He was hospitalized and underwent surgical decompression and stabilization followed by immobilization in a halo brace for 3 months. He has regained most motor and sensory function. His temperature is 37°C (98.6°F), pulse is 70/min and regular, and blood pressure is 130/70 mm Hg. Cranial nerves are intact. Muscle strength is 4/5 in the deltoid and biceps muscles bilaterally and is 5/5 otherwise. Deep tendon reflexes are 1+ bilaterally in the biceps and 3+ bilaterally in the triceps, quadriceps, and gastrocnemius muscles. Babinski sign is present bilaterally. Sensation to pinprick and temperature is mildly decreased below the upper chest. Which of the following is the most likely cause of this patient's urinary symptoms?

- A) Detrusor hyperreflexia
- B) Diabetes insipidus
- C) Flaccid bladder
- D) Overflow incontinence
- E) Outlet obstruction

19. A previously healthy 25-year-old woman is brought to the emergency department 30 minutes after being struck by a motor vehicle. On arrival, she is alert and oriented and reports leg pain. Her vital signs are within normal limits. Examination shows normal sensory and motor function in the foot. Posterior tibial and dorsalis pedis pulses are normal. X-rays show a minimally displaced comminuted midshaft fracture of the tibia. The lower extremity is splinted in the emergency department, and the patient is admitted for observation. Three hours later, she has numbness in her foot and increasing pain in her lower extremity. Examination now shows decreased sensation to light touch over the foot and pain with passive extension of the toes. Which of the following is the most appropriate next step in management?

- ☐ A) Elevation of the extremity
- ☐ B) Measurement of compartment pressures
- ☐ C) Anticoagulant therapy
- ☐ D) Angiography
- ☐ E) Closed reduction
- ☐ F) Nerve exploration
- ☐ G) Open reduction only
- ☐ H) Open reduction and internal fixation

20. A 57-year-old man comes to the physician because of a 2-week history of constant headache, abdominal pain, and dizziness. He has had generalized weakness and decreased vision during this period. He has no history of serious illness or visual problems and takes no medications. He has lived and worked on a rural farm all of his life. He drinks 4 oz of homemade liquor five times weekly. His blood pressure is 135/92 mm Hg. The pupils are 3 mm and react slowly to bright light. The patient is unable to read any letters on the Snellen chart but can count fingers held 1 foot in front of his face. Funduscopic examination shows pallor of the optic discs. Abdominal examination shows diffuse tenderness but no rebound. Muscle strength is 4/5 in all extremities. Exposure to which of the following toxins is the most likely cause of this patient's symptoms?

- ☐ A) n-Hexane
- ☐ B) Methanol
- ☐ C) Methyl ethyl ketone
- ☐ D) Organophosphate
- ☐ E) Toluene
- ☐ F) Trichloroethylene

☐ Mark

21. A previously healthy 72-year-old woman is brought to the physician by her daughter because of progressive memory loss and confusion during the past 18 months. Although she has had mild forgetfulness for 5 years, her symptoms dramatically increased during the past 18 months. An S₄ gallop is heard. There is a right carotid bruit. Peripheral pulses are decreased. Babinski sign is present on the left. Which of the following is the most appropriate pharmacotherapy?

- ☐ A) Aspirin
- ☐ B) Clopidogrel
- ☐ C) Dipyridamole
- ☐ D) Heparin
- ☐ E) Warfarin

22. An 18-year-old man comes to the physician because of difficulty walking for 1 week. Examination shows visual fields that are constricted equally in both eyes on close and distant testing. He has collapsing effort on muscle strength testing in the left extremities. He staggers from side to side when walking but does not fall. On Romberg testing, he abruptly falls to the left. There is hemisensory loss to light touch, pinprick, vibration, and proprioception on the left that stops exactly at the midline. Which of the following is the most likely diagnosis?

- ☐ A) Cerebellar degeneration
- ☐ B) Cervical spinal cord compression
- ☐ C) Conversion disorder
- ☐ D) Guillain-Barré syndrome
- ☐ E) Multiple sclerosis
- ☐ F) Sensory neuropathy
- ☐ G) Tabes dorsalis
- ☐ H) Vitamin B₁₂ (cobalamin) deficiency

☐ Mark

23. A 50-year-old man has had low back pain radiating to the right buttock and posterior thigh to the knee since lifting a 50-lb weight 18 hours ago. He has had urinary incontinence during this time. The Achilles tendon reflex is absent on the right, and there is decreased perianal sensation. Straight-leg raising causes a marked increase in pain in the posterior thigh and leg. Which of the following is the most likely cause of these findings?

- ☐ A) Cauda equina compression
- ☐ B) Epidural hematoma
- ☐ C) Lumbosacral strain
- ☐ D) S1 nerve root radiculopathy
- ☐ E) Transverse myelitis

☐ Mark

24. An 87-year-old woman is brought to the physician by her son because of a 7-month history of decreasing ability to function independently. She is no longer able to write checks and often gets lost while driving. She has hypertension and hyperlipidemia. Current medications include chlorothiazide and pravastatin. Her temperature is 37°C (98.6°F), pulse is 84/min, respirations are 12/min, and blood pressure is 170/100 mm Hg. Physical examination shows no other abnormalities. Mental status examination shows poor concentration. She is oriented to person and place but not to time. She is able to recall zero out of three objects after 5 minutes. Cognition and constructional ability are decreased. Laboratory findings are within the reference range. Which of the following is the most appropriate next step in diagnosis?

- ☐ A) Measurement of serum ammonia concentration
- ☐ B) Neuropsychological testing
- ☐ C) PET scan
- ☐ D) MRI of the brain
- ☐ E) EEG

25. A 32-year-old woman is brought to the emergency department 1 hour after the acute onset of left-sided weakness involving the face, arm, and leg. She had rheumatic fever during childhood and was told at that time that she had a heart murmur. Her pulse is 92/min and irregularly irregular. Examination shows moderate left-sided weakness. Babinski sign is present on the left. Which of the following is the most likely cause of these findings?

- ☐ A) Atrial fibrillation
- ☐ B) Bacterial endocarditis
- ☐ C) Carotid artery dissection
- ☐ D) Carotid atherosclerosis
- ☐ E) Diabetes mellitus
- ☐ F) Hypertension
- ☐ G) Meningovascular syphilis
- ☐ H) Migraine
- ☐ I) Mitral valve prolapse
- ☐ J) Patent foramen ovale
- ☐ K) Subclavian steal syndrome
- ☐ L) Ventricular septal defect

26. A 42-year-old man is brought to the emergency department 20 minutes after the onset of generalized tonic-clonic seizures. He has a 3-year history of a seizure disorder treated with phenytoin, but he does not take his medication as prescribed. He has long-standing alcoholism. He is stuporous. Neurologic examination shows no focal findings. His hemoglobin concentration is 8.8 g/dL, and mean corpuscular volume is 105 μm^3 . A deficiency of which of the following is the most likely cause of this patient's hematologic findings?

- ☐ A) Folic acid
- ☐ B) Vitamin B₁ (thiamine)
- ☐ C) Vitamin B₂ (riboflavin)
- ☐ D) Vitamin B₆
- ☐ E) Vitamin B₁₂ (cobalamin)
- ☐ F) Vitamin C

27. A 12-year-old girl is brought to the physician by her parents because of a 2-week history of sudden, uncontrollable movements of her face, arms, and legs. When she attempts to control these movements, they increase in frequency and intensity. Her symptoms do not occur during sleep. She has not had loss of consciousness. Two months ago, she had a sore throat that resolved without treatment. Her parents report that she is well behaved, performs well in school, and has not had any recent stressors. Her temperature is 37.5°C (99.5°F), pulse is 90/min, respirations are 20/min, and blood pressure is 80/40 mm Hg. Examination shows frequent irregular, spasmodic movements of the face and upper and lower extremities. The remainder of the examination shows no abnormalities. A CT scan of the head and an EEG show no abnormalities. Which of the following is the most likely diagnosis?

- ☐ A) Absence seizures
- ☐ B) Complex partial seizures
- ☐ C) Conversion disorder
- ☐ D) Sydenham chorea
- ☐ E) Tourette disorder

28. A previously healthy 72-year-old man is brought to the physician 1 day after the sudden loss of vision in his left eye that resolved within 45 minutes. He had a similar episode in the same eye 2 weeks ago. He has not had headache, fatigue, weakness, jaw pain, or difficulty breathing. He appears well. His pulse is 72/min, respirations are 18/min, and blood pressure is 172/85 mm Hg. Examination shows no abnormalities. A grade 2/6, early systolic murmur is heard with occasional ectopic beats. Which of the following is the most appropriate next step in diagnosis?

- ☐ A) Slit-lamp examination of the left eye
- ☐ B) Determination of erythrocyte sedimentation rate
- ☐ C) 24-Hour ambulatory ECG monitoring
- ☐ D) Carotid duplex ultrasonography
- ☐ E) Echocardiography
- ☐ F) CT scan of the head

29. An 82-year-old woman with dementia, Alzheimer type, is brought to the physician by her husband because he has had progressive difficulty feeding her over the past 4 weeks. He says she no longer swallows food that he puts in her mouth, even with prompting. She no longer says his name, and he believes that she does not recognize him. He has been her primary caregiver for 4 years. She does not have an advance directive. He says he does not want to hasten her death. He wants to continue to take care of her at home but is concerned about his ability to do so. He declines placement of a gastrostomy tube. She is 163 cm (5 ft 4 in) tall and weighs 44 kg (98 lb); BMI is 17 kg/m². She appears well groomed but malnourished. She is not oriented to person, place, or time. The remainder of the examination shows no abnormalities. Which of the following is the most appropriate recommendation?

- ☐ A) Oral multivitamin supplementation at home
- ☐ B) Oral nutritional supplementation at home
- ☐ C) Hospice care at home
- ☐ D) Admission to the hospital for total parenteral nutrition
- ☐ E) Transfer to a skilled nursing care facility for tube feeding

30. A 47-year-old man comes to the physician because of a 10-day history of drooping of the right eyelid. He has no history of serious illness and takes no medications. Vital signs are within normal limits. Physical examination shows anhidrosis over the right side of the face. The right eyelid droops 3 mm. The right pupil is 3 mm, and the left pupil is 5 mm; both pupils are reactive. Visual acuity is 20/30 on the right and 20/25 on the left. Cranial nerves are otherwise intact. Muscle strength is 5/5 throughout. Deep tendon reflexes are 1+ throughout. Sensation is intact. An MRI of which of the following is most likely to confirm the diagnosis?

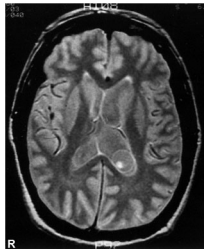
- ☐ A) Lower neck
- ☐ B) Medulla
- ☐ C) Midbrain
- ☐ D) Ocular orbit
- ☐ E) Pons

31. A 52-year-old man comes to the physician because of a 1-month history of numbness and tingling of the ring and small fingers of his right hand that radiates to his palm. His symptoms are now constant and exacerbated at night. He has a 2-week history of decreased grip strength and wasting of the small muscles in his right hand. He has not had neck or arm pain. Four months ago, he was diagnosed with pancreatic cancer; since then, he underwent Whipple procedure and is currently receiving chemotherapy with gemcitabine. Since his diagnosis, he has had significant fatigue and an 18-kg (40-lb) weight loss. He is 173 cm (5 ft 8 in) tall and now weighs 54 kg (120 lb); BMI is 18 kg/m². His temperature is 37°C (98.6°F), pulse is 95/min, and blood pressure is 120/79 mm Hg. There is wasting of the interosseous muscles on the right. Muscle strength is 3/5 in the right interosseous and hypothenar muscles and flexor digitorum profundus of the ring and small fingers; it is full elsewhere. Deep tendon reflexes are 2+ throughout. Babinski sign is absent. Sensation to pinprick and temperature is decreased over the ulnar aspect of the palm and dorsum of the hand and ring and small fingers; proprioception is intact. The remainder of the neurologic examination shows no abnormalities. Which of the following is the most likely cause of this patient's symptoms?

- ☐ A) C8 nerve root infiltration by tumor
- ☐ B) Cervical disc herniation
- ☐ C) Chemotherapy neurotoxicity
- ☐ D) Ulnar nerve compression
- ☐ E) Vitamin B₁₂ (cobalamin) deficiency

32. A 47-year-old man who recently emigrated from Mexico is brought to the emergency department 1 hour after a generalized tonic-clonic seizure that lasted 3 minutes. He has a 3-week history of constant headaches. He is otherwise healthy. He is sexually active. He has smoked one pack of cigarettes daily for 22 years. Physical and neurologic examinations show no abnormalities. An MRI of the brain is shown. Which of the following is the most likely risk factor for the development of this patient's condition?

- ☐ A) Airborne infection
- ☐ B) Arthropod-borne infection
- ☐ C) Fecal contamination of food
- ☐ D) Sexual intercourse with an infected partner
- ☐ E) Smoking tobacco



33. An 82-year-old woman is brought to the emergency department because of an 8-hour history of fever and progressive confusion. Four days ago, she attended a picnic and ate barbecued hot dogs and homemade unpasteurized cheese. She has severe pemphigus treated with daily oral prednisone. On arrival, she is somnolent. Her temperature is 39.4°C (103°F). Physical examination shows a rigid neck. Neurologic examination shows no focal findings. A lumbar puncture yields turbid fluid. Cerebrospinal fluid analysis shows:

Glucose	24 mg/dL
Protein	180 mg/dL
Leukocyte count	260/mm ³
Segmented neutrophils	60%
Monocytes	40%

A Gram stain of cerebrospinal fluid shows numerous gram-positive bacilli. Which of the following is the most likely causal organism?

- ☐ A) *Bacillus subtilis*
- ☐ B) *Clostridium perfringens*
- ☐ C) *Corynebacterium* species
- ☐ D) *Haemophilus influenzae*
- ☐ E) *Listeria monocytogenes*
- ☐ F) *Streptococcus pneumoniae*

34. A 47-year-old man is brought to the emergency department by his friends 30 minutes after a generalized tonic-clonic seizure that lasted 2 minutes. He had been smoking crack cocaine when the seizure began. He is unresponsive. Examination shows a 7-mm dilated left pupil that is nonreactive to light. There is drooping of the right lower face. He has intermittent decerebrate posturing in the right upper and lower extremities. Deep tendon reflexes are hyperactive on the right. Babinski sign is present on the right. A CT scan of the head shows an intracerebral hemorrhage. Which of the following types of cerebral herniation is the most likely cause of his condition?

- Ⓐ A) Central
- Ⓑ B) Cerebellar tonsillar
- Ⓒ C) Subfalcine
- Ⓓ D) Transtentorial
- Ⓔ E) Upward

35. A 27-year-old woman comes to the physician because of an 18-hour history of progressive loss of vision and pain with movement of the right eye. Her visual acuity is 20/200 in the right eye and 20/20 in the left eye. The pupils are equal; the right pupil reacts poorly to direct light but reacts consensually to light in the opposite pupil. Funduscopic examination shows normal findings. At a follow-up visit 8 weeks later, vision in the right eye is 20/30. There is mild pallor of the right optic disc. The remainder of the examination is unchanged. This patient is at greatest risk for which of the following?

- A) Cerebral infarction
- B) Lyme disease
- C) Multiple sclerosis
- D) Systemic lupus erythematosus
- E) Wegener granulomatosis

36. A 79-year-old woman comes to the physician because of a 4-month history of severe headaches located in the right temporal area and pain in the jaw when chewing. Over the past 2 months, she has had diffuse pain in the back, shoulders, and extremities that is worse on mild exertion. There is tenderness to palpation over the temporal muscles bilaterally. Neurologic examination shows no focal findings. Which of the following is the most likely cause of this patient's limb pain?

- A) Chronic fatigue syndrome
- B) Fibromyalgia
- C) Lyme disease
- D) Polymyalgia rheumatica
- E) Rheumatoid arthritis
- F) Systemic lupus erythematosus

37. A 2-year-old girl has the sudden onset of unsteady gait following chickenpox. Physical examination shows an irritable but alert child with normal vital signs. She is unable to stand or sit unsupported. There is bilateral horizontal nystagmus. The neck is supple. Analysis of cerebrospinal fluid shows 14 leukocytes/mm³ (N=0–7) and a protein concentration of 60 mg/dL. Gram stain shows no organisms. Which of the following is the most likely diagnosis?

- A) Acute cerebellar ataxia
- B) Ataxia-telangiectasia
- C) Benign paroxysmal vertigo
- D) Cerebellar astrocytoma
- E) Friedreich ataxia

38. A 67-year-old woman with rheumatoid arthritis comes to the physician because of a 2-month history of progressive weakness of her hands. During this time, she has had a tingling sensation in her hands and has had increasing difficulty with everyday tasks, such as buttoning a shirt or writing. Her rheumatoid arthritis symptoms have been managed with corticosteroid therapy for 10 years. Examination shows marked rheumatoid deformities of the fingers and weakness of finger extensors and wrist flexors bilaterally. Deep tendon reflexes are 3+ in the upper extremities. Which of the following is the most appropriate next step in diagnosis?

- A) Measurement of serum creatine kinase activity
- B) X-rays of the hands
- C) DEXA scan
- D) MRI of the cervical spine
- E) Electromyography and nerve conduction studies



39. A 27-year-old woman is brought to the physician because of an 8-week history of double vision. She says that the double vision most often occurs when she looks upward and to the right. She is often asymptomatic in the morning. Her friends have commented on droopiness of both eyelids. Her only medication is an oral contraceptive. Examination shows bilateral ptosis that is greater in the right eye than in the left. The ptosis is worsened by gazing upward for 60 seconds. There is weakness of abduction of the right eye and mild weakness of the frontal muscle. The remainder of the neurologic examination shows no abnormalities. Which of the following is the most appropriate initial pharmacotherapy for this patient?

- A) Azathioprine
- B) Cyclosporine
- C) Immune globulin
- D) Prednisone
- E) Pyridostigmine

40. A 70-year-old man is brought to the emergency department 1 hour after the sudden onset of incoordination and clumsiness; he has fallen several times. Physical examination shows a broad-based unsteady gait and a resting fine tremor of the hands. His left eye does not move past midline. He is oriented to person but not to place or time and is vague and rambling in his history. Which of the following is the most likely diagnosis?

- A) Friedreich ataxia
- B) Hepatolenticular degeneration (Wilson disease)
- C) Spinocerebellar ataxia, type I
- D) Wernicke encephalopathy



41. A 57-year-old woman comes to the physician because of a 3-month history of progressive slurred speech and gait unsteadiness. She also has had mild occipital headache each morning for 3 weeks. She has type 2 diabetes mellitus well controlled with diet. She takes no medications. Examination shows atrophy and fasciculation of the right side of the tongue. The tongue deviates to the right on protrusion. Other cranial nerves are intact. Her gait is stiff with a normal base. Muscle strength is 4/5 in the triceps, wrist extensors, interossei, and iliopsoas muscles bilaterally. Deep tendon reflexes are 3+ diffusely. Babinski sign is present bilaterally. Sensory examination shows no abnormalities. Which of the following is the most likely diagnosis?

- A) Epidural abscess
- B) Epidural spinal cord metastasis
- C) Syringomyelia
- D) Tumor of the foramen magnum
- E) Vertebral osteomyelitis

42. A 24-year-old woman has weakness of her right leg 8 hours after a prolonged but uncomplicated vaginal delivery of a 3629-g (8-lb) full-term newborn. Examination shows weakness of dorsiflexion and eversion of the right ankle; inversion and plantar flexion are normal. There is an area of numbness on the dorsum of the foot. Reflexes are intact. Which of the following is the most likely location of the lesion?

- | | |
|--------------------------|--------------------------|
| A) Basal ganglia | H) Frontoparietal cortex |
| B) Brachial plexus | I) Internal capsule |
| C) Brain stem | J) Lumbar nerve root |
| D) Cervical nerve root | K) Lumbosacral plexus |
| E) Common peroneal nerve | L) Median nerve |
| F) Conus medullaris | M) Muscle |
| G) Femoral nerve | N) Sciatic nerve |



43. A 25-year-old woman has had daily frontal headaches for 2 months. The pain generally begins as a dull ache soon after waking and gradually worsens. The pain is exacerbated by loud noise or stress and relieved by acetaminophen and sleep. Physical examination shows no abnormalities. Which of the following is the most likely diagnosis?

- A) Cerebral neoplasm
- B) Cluster headache
- C) Frontal sinusitis
- D) Migraine with aura
- E) Migraine without aura
- F) Temporal arteritis
- G) Tension-type headache
- H) Trigeminal neuralgia

44. A 74-year-old woman is brought to the emergency department 1 hour after the sudden onset of partial blindness. She has had low-grade fever; occipital headaches; pain of the shoulders, pelvic area, and joints; and morning stiffness for 2 months. Her temperature is 37.7°C (99.8°F), and blood pressure is 138/88 mm Hg. There is generalized muscle tenderness but no weakness or atrophy. Ophthalmologic examination shows evidence of a right central retinal artery occlusion. Which of the following is the most likely diagnosis?

- A) Fibromyositis
- B) Polyarteritis nodosa
- C) Polymyositis
- D) Systemic lupus erythematosus
- E) Temporal arteritis

45. A 52-year-old man comes to the physician because of a 2-hour history of a diffuse, bilateral, dull headache. He has hypertension and has been poorly compliant with his medication. He is otherwise asymptomatic. He is alert but confused. His blood pressure is 240/150 mm Hg. Funduscopic examination shows hemorrhages and exudates. Which of the following is the most appropriate pharmacotherapy?

- | | |
|------------------|------------------|
| A) Amitriptyline | G) Nifedipine |
| B) Clonidine | H) Nitroprusside |
| C) Dexamethasone | I) Prednisone |
| D) Ergotamine | J) Propranolol |
| E) Ibuprofen | K) Sumatriptan |
| F) Methysergide | |



46. A 20-year-old man is comatose 3 hours after being hit behind the left ear with a bat while catching in a softball game. He was dazed for a few seconds after the injury but was lucid on arrival in the emergency department 2 hours ago. Examination at that time showed a large bruise behind the left ear, no other abnormalities were noted. Which of the following is the most likely explanation for these findings?

- A) Electrophysiologic dysfunction
- B) Epidural hemorrhage
- C) Hemorrhage into the cerebral hemisphere
- D) Shearing and disruption of axons
- E) Subdural hemorrhage

47. A 67-year-old woman comes to the physician because of a 4-year history of gradually progressive shaking of her head and hands. The shaking decreases after she drinks a glass of wine. She is right-hand dominant. She has a history of hypertension treated with benazepril and hyperlipidemia treated with atorvastatin. She has consumed four glasses of wine daily for 35 years. Her blood pressure is 142/92 mm Hg. Examination shows a rapid tremor of the head and voice. There is a high-frequency tremor of the hands on finger-nose testing. Which of the following is the most appropriate pharmacotherapy for this patient's tremor?

- Ⓐ A) Carbidopa-levodopa
- Ⓑ B) Disulfiram
- Ⓒ C) Phenytoin
- Ⓓ D) Propranolol
- Ⓔ E) Trihexyphenidyl

48. A 32-year-old woman comes to the physician because of a 6-week history of difficulty falling asleep and early morning awakening. She says that she is tired throughout the day and has become increasingly irritable with her family and coworkers. She finds it difficult to go to work. She has not noticed any improvement in her sleep despite not drinking coffee past noon or exercising in the evening. She no longer enjoys gardening and teaching her children to make crafts. She feels guilty about how her sleep deprivation is affecting her family. She has had a decreased appetite resulting in a 4.5-kg (10-lb) weight loss during the past 6 weeks. She has no history of psychiatric illness. She takes no medications. She does not smoke or use illicit drugs and only occasionally drinks alcohol. She appears tired. Her temperature is 37°C (98.6°F), pulse is 80/min, and blood pressure is 118/74 mm Hg. Physical examination shows no abnormalities. Mental status examination shows a sad affect. She becomes tearful and says that she has been sad since her father was diagnosed with lung cancer 4 months ago. She says, "I have thought about killing myself, but I would never hurt my children in that way." In addition to psychotherapy, which of the following is the most appropriate initial step in management?

- A) Support group for relatives of cancer victims
- B) Buspirone therapy
- C) Diphenhydramine therapy at bedtime
- D) Fluoxetine therapy
- E) Lorazepam therapy

49. A 27-year-old man comes to the emergency department because he has been unable to void for 24 hours. Over the past 6 days, he has had gradually progressive bilateral leg weakness. He had the onset of low back pain 7 days ago while working in the garden. He has a history of intravenous drug use. His temperature is 38.9°C (102°F). Examination shows moderate weakness and hyperreflexia of the lower extremities bilaterally. Babinski sign is present bilaterally. Sensation to pinprick is decreased below the umbilicus, and sensation to light touch, pinprick, proprioception, and vibration is decreased over both lower extremities. Which of the following is the most likely location of the lesion?

- ☐ A) Cervical spinal cord
- ☐ B) Conus medullaris
- ☐ C) Femoral nerve
- ☐ D) Lumbar nerve root
- ☐ E) Lumbosacral plexus
- ☐ F) Neuromuscular junction
- ☐ G) Sciatic nerve
- ☐ H) Thoracic spinal cord
- ☐ I) Ulnar nerve

☐ Mark

50. A 67-year-old woman is brought to the emergency department by her husband 1 hour after the sudden onset of incomprehensible speech that began shortly after breakfast. Her husband describes her speech during the episode as fluent and well articulated but full of illogical word combinations, making it impossible for her to be understood. He says that she also did not seem to understand him or follow commands. She has hypertension, hyperlipidemia, and asthma. Current medications include lovastatin and lisinopril. She is allergic to aspirin. On arrival, her language function is normal. Her pulse is 80/min and regular, and blood pressure is 152/88 mm Hg. Physical and neurologic examinations show no abnormalities. Echocardiography shows no abnormalities. Carotid duplex ultrasonography shows 30% stenosis of the left common carotid artery; the right is normal. Which of the following is the most appropriate intervention to decrease this patient's risk for future episodes?

- A) Clopidogrel therapy
- B) Estrogen replacement therapy
- C) Propranolol therapy
- D) Warfarin therapy
- E) Left carotid endarterectomy